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EV Charger Socket Locking Design

Abstract

An EV charger locking system includes a plunger, one or more springs, and an electrically controlled actuator. The plunger contacts a lockable portion of an EV charger inserted into a channel of an EV charging socket. When uncompressed, the one or more springs position a tip of the plunger in a return position in or above a hole of the channel. When compressed, the one or more springs apply a force in a second direction to return the tip to the return position. When activated, the actuator applies a mechanical force to the plunger in a first direction to move the tip through the hole and to position the tip in a locking position in the channel to contact the lockable portion. When deactivated, the actuator removes the mechanical force from the plunger, allowing the plunger to be returned to the return position by the one or more springs.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/559,340, filed on Feb. 29, 2024, the content of which is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

[0002] The teachings herein relate to a locking mechanism in an electric vehicle (EV) charging socket. More particularly the teachings herein relate to apparatus and methods for automatically locking an EV charger in an EV charging socket.

BACKGROUND

[0003] There are numerous reports of EV chargers falling out of charging sockets or becoming disengaged from charging sockets while charging. There are also reports of EV chargers becoming stuck in or jamming charging sockets after charging is completed.

[0004] EV charger disengagement is a significant problem due to the long charging times required for EVs and the fact that, due to the long charging times, EV chargers are often left unattended. As a result, any disengagement of an EV charger from a charging socket can result in a significant delay in charging an EV.

[0005] EV charger jamming is also a significant problem because it reduces the availability of an already limited amount of EV chargers available. A jammed EV charger connection not only inconveniences the current person charging their EV, it also prevents others from using that EV charger.

[0006] EV charger disengagement or jamming can occur for many different reasons. For example, due to normal wear and tear, an EV charger may become loose in a socket. Consequently, the charger or its locking mechanism may no longer fit snugly and connect properly with a socket. Alternatively, the loose fit may allow the charger to be misaligned causing it to jam or get stuck in the socket.

[0007] User error may also contribute to EV charger disengagement or jamming in a socket. A user may not properly insert and align an EV charger in a socket. As a result, the EV charger may disengage from the socket or even fall out during charging. Alternatively, a user may apply too much pressure to a misaligned engagement causing the EV charger to jam.

[0008] The environment of the charger may also contribute to EV charger disengagement or jamming. In certain climates, the charger may be exposed to excessive humidity, dust, or corrosive materials. This may cause the charger or its components to change shape leading to connection problems with a socket or leading to the charger becoming stuck in a socket.

[0009] Finally, EV chargers come in all shapes and sizes. Some have long cables and some have short cables. Some have thick cables, while others have thin cables. These varied characteristics can contribute to EV charger disengagement or jamming. For example, a connector with a long thick cable is likely to place more force on a connection between the EV charger and socket and result in the EV charger disengaging or jamming.

[0010] Consequently, there is an unmet need for systems and methods that can prevent EV chargers from disengaging from sockets during charging and prevent them from jamming or getting stuck in sockets after charging.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The skilled artisan will understand that the drawings, described below, are for illustration purposes only. The drawings are not intended to limit the scope of the present teachings in any way.

[0012] FIG. **1** is a block diagram that illustrates a computer system, upon which embodiments of the present teachings may be implemented.

[0013] FIG. **2** is an exemplary three-dimensional diagram showing how an electrically controlled locking mechanism housing is connected to an EV charging socket, in accordance with various embodiments.

[0014] FIG. **3** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism is connected internally to an EV charging socket, in accordance with various embodiments.

[0015] FIG. **4** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism connected to an EV charging socket locks an EV charger during charging, in accordance with various embodiments.

[0016] FIG. **5** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism connected to an EV charging socket unlocks an EV charger after charging, in accordance with various embodiments.

[0017] FIG. **6** is a series of exemplary cross-sectional diagrams showing how an electrically controlled locking mechanism connected to an EV charging socket prevents an EV charger from becoming jammed in the EV charging socket when removal of the EV charger is attempted before the electrically controlled locking mechanism is completely disengaged after charging, in accordance with various embodiments.

[0018] FIG. **7** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator is connected internally to an EV charging socket, in accordance with various embodiments.

[0019] FIG. **8** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket locks an EV charger during charging, in accordance with various embodiments.

[0020] FIG. **9** is an exemplary cross-sectional diagram showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket unlocks an EV charger after charging, in accordance with various embodiments.

[0021] FIG. **10** is series of exemplary cross-sectional diagrams showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket prevents an EV charger from becoming jammed in the EV charging socket when removal of the EV charger is attempted before the electrically controlled locking mechanism is completely disengaged, in accordance with various embodiments.

[0022] FIG. **11** is an exemplary flowchart showing a method for locking an EV charger in an EV charging socket, in accordance with various embodiments.

[0023] Before one or more embodiments of the present teachings are described in detail, one skilled in the art will appreciate that the present teachings are not limited in their application to the details of construction, the arrangements of components, and the arrangement of steps set forth in the following detailed description or illustrated in the drawings. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DESCRIPTION OF VARIOUS EMBODIMENTS

Computer-Implemented System

[0024] FIG. **1** is a block diagram that illustrates a computer system **100**, upon which embodiments of the present teachings may be implemented. Computer system **100** includes a bus **102** or other communication mechanism for communicating information, and a processor **104** coupled with bus

102 for processing information. Computer system **100** also includes a memory **106**, which can be a random-access memory (RAM) or other dynamic storage device, coupled to bus **102** for storing instructions to be executed by processor **104**. Memory **106** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **104**. Computer system **100** further includes a read only memory (ROM) **108** or other static storage device coupled to bus **102** for storing static information and instructions for processor **104**. A storage device **110**, such as a magnetic disk or optical disk, is provided and coupled to bus **102** for storing information and instructions.

[0025] Computer system **100** may be coupled via bus **102** to a display **112**, such as a cathode ray tube (CRT) or liquid crystal display (LCD), for displaying information to a computer user. An input device **114**, including alphanumeric and other keys, is coupled to bus **102** for communicating information and command selections to processor **104**. Another type of user input device is cursor control **116**, such as a mouse, a trackball or cursor direction keys for communicating direction information and command selections to processor **104** and for controlling cursor movement on display **112**.

[0026] A computer system **100** can perform the present teachings. Consistent with certain implementations of the present teachings, results are provided by computer system **100** in response to processor **104** executing one or more sequences of one or more instructions contained in memory **106**. Such instructions may be read into memory **106** from another computer-readable medium, such as storage device **110**. Execution of the sequences of instructions contained in memory **106** causes processor **104** to perform the process described herein.

[0027] Alternatively, hard-wired circuitry may be used in place of or in combination with software instructions to implement the present teachings. For example, the present teachings may also be implemented with programmable artificial intelligence (AI) chips with only the encoder neural network programmed—to allow for performance and decreased cost. Thus, implementations of the present teachings are not limited to any specific combination of hardware circuitry and software.

[0028] The term “computer-readable medium” or “computer program product” as used herein refers to any media that participates in providing instructions to processor **104** for execution. The terms “computer-readable medium” and “computer program product” are used interchangeably throughout this written description. Such a medium may take many forms, including but not limited to, non-volatile media and volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **110**. Volatile media includes dynamic memory, such as memory **106**.

[0029] Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, or any other magnetic medium, a CD-ROM, digital video disc (DVD), a Blu-ray Disc, any other optical medium, a thumb drive, a memory card, a RAM, PROM, and EPROM, a FLASH-EPROM, any other memory chip or cartridge, or any other tangible medium from which a computer can read.

[0030] Various forms of computer readable media may be involved in carrying one or more sequences of one or more instructions to processor **104** for execution. For example, the instructions may initially be carried on the magnetic disk of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a telephone line using a modem. A modem local to computer system **100** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector coupled to bus **102** can receive the data carried in the infra-red signal and place the data on bus **102**. Bus **102** carries the data to memory **106**, from which processor **104** retrieves and executes the instructions. The instructions received by memory **106** may optionally be stored on storage device **110** either before or after execution by processor **104**.

[0031] In accordance with various embodiments, instructions configured to be executed by a processor to perform a method are stored on a computer-readable medium.

[0032] The computer-readable medium can be a device that stores digital information. The computer-readable medium is accessed by a processor suitable for executing instructions configured to be executed.

[0033] The following descriptions of various implementations of the present teachings have been presented for purposes of illustration and description. It is not exhaustive and does not limit the present teachings to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practicing of the present teachings. Additionally, the described implementation includes software but the present teachings may be implemented as a combination of hardware and software or in hardware alone. The present teachings may be implemented with both object-oriented and non-object-oriented programming systems.

Electrically Controlled EV Charger Lock

[0034] As described above, there are numerous reports of EV chargers falling out of charging sockets or becoming disengaged from charging sockets while charging. There are also reports of EV chargers becoming stuck in or jamming charging sockets after charging is completed.

[0035] EV charger disengagement is a significant problem due to the long charging times required for EVs and the fact that, due to the long charging times, EV chargers are often left unattended. As a result, any disengagement of an EV charger from a charging socket can result in a significant delay in charging an EV.

[0036] EV charger jamming is also a significant problem because it reduces the availability of an already limited amount of EV chargers available. A jammed EV charger connection not only inconveniences the current person charging their EV, it also prevents others from using that EV charger.

[0037] EV charger disengagement or jamming can occur for many different reasons, including, but not limited to, normal wear and tear, user error, environmental factors, and the varied characteristics of EV chargers.

[0038] In various embodiments, EV charger disengagement or jamming is prevented by modifying an EV charging socket to include an automatic locking mechanism for locking an inserted EV charger to the EV charging socket. The locking mechanism is made to be automatic by placing it under electrical control. An automatic or electrically controlled locking mechanism is less susceptible to EV charger disengagement or jamming. This is because an automatic or electrically controlled locking mechanism is less susceptible to, for example, normal wear and tear, user error, environmental factors, and the varied characteristics of EV chargers.

[0039] FIG. 2 is an exemplary three-dimensional diagram **200** showing how an electrically controlled locking mechanism housing is connected to an EV charging socket, in accordance with various embodiments. In FIG. 2, electrically controlled locking mechanism housing **220** is shown connected to EV charging socket **210** beyond the opening of EV charging socket **210** in order to be able to engage the lockable portion of an inserted EV charger (not shown).

[0040] FIG. 3 is an exemplary cross-sectional diagram **300** showing how an electrically controlled locking mechanism is connected internally to an EV charging socket, in accordance with various embodiments. In FIG. 3, electrically controlled locking mechanism housing **220** is connected to the outside of EV charging socket **210**. Electrically controlled locking mechanism housing **220** includes the electrically controlled locking mechanism.

[0041] The electrically controlled locking mechanism includes electrically controlled actuator **221**, plunger or lock pin **222**, and one or more springs **223**. Electrically controlled actuator **221** is, for example, an electromagnetic actuator or solenoid. EV charging socket **210** includes a hole **211** into channel **212**. Hole **211** is located in a wall of channel **212** beyond opening **213** into which an EV charger can be inserted. The electrically controlled locking mechanism is positioned above hole **211** so that plunger **222** can be extended through hole **211** and into channel **212** of EV charging socket **210**.

[0042] In various embodiments, plunger **222** is not always connected to shaft **224** of actuator **221**.

For example, shaft **224** may only be physically connected to plunger **222** when actuator **221** is activated. As shown in FIG. **3**, actuator **221** is not activated. Consequently, plunger **222** is not in contact with shaft **224**, and one or more springs **223** cause plunger **222** to be pushed upward through hole **211** so that plunger **222** does not affect the insertion or removal of an EV charger into or out of channel **212**.

[0043] In various embodiments, electrically controlled actuator **221** is activated or deactivated using electrical circuitry **240**. Electrical circuitry **240** detects when the EV charger is charging or not charging. For example, electrical circuitry **240** can be electrically connected to the terminals of EV charging socket **210** and, therefore, can determine whether or not electric current is being supplied to the terminals of EV charging socket **210**.

[0044] In various embodiments, electrical circuitry **240** can include one or more processors. The one or more processors can be, but are not limited to, controllers, computers, microprocessors, computer systems such as the computer system of FIG. **1**, or any devices capable of sending and receiving control signals or data and analyzing data. In other words, electrically controlled actuator **221** can be activated or deactivated by receiving controlled signals from one or more processors.

[0045] FIG. **4** is an exemplary cross-sectional diagram **400** showing how an electrically controlled locking mechanism connected to an EV charging socket locks an EV charger during charging, in accordance with various embodiments. In FIG. **4**, EV charger **230** has been inserted into the channel of EV charging socket **210**. EV charger **230** includes lockable portion or notch **231**.

[0046] To lock EV charger **230** to or in EV charging socket **210**, electrically controlled actuator **221** of the electrically controlled locking mechanism is activated. This causes electrically controlled actuator **221** to extend shaft **224** downward making contact with and pushing plunger **222** downward also. This downward movement of plunger **222** causes one or more springs **223** to be compressed. It also causes plunger **222** to move down through hole **211** of EV charging socket **210**, to engage notch **231** of EV charger **230**, and to lock EV charger **230** to EV charging socket **210**. Note that actuator **221** remains activated throughout charging in order to keep EV charger **230** locked to EV charging socket **210**.

[0047] FIG. **5** is an exemplary cross-sectional diagram **500** showing how an electrically controlled locking mechanism connected to an EV charging socket unlocks an EV charger after charging, in accordance with various embodiments. In FIG. **5**, electrically controlled actuator **221** has been deactivated in order to unlock EV charger **230**.

[0048] The deactivation of actuator **221** causes shaft **224** to move upward and remove the pressure on plunger **222**. In turn, one or more springs **223** cause plunger **222** to also move upward and out of notch **231** of EV charger **230**. The movement of plunger **222** upward through hole **211** and out of the channel of EV charging socket **210** unlocks EV charger **230** from EV charging socket **210** and allows EV charger **230** to be removed from socket **210**.

[0049] FIG. **6** is series **600** of exemplary cross-sectional diagrams showing how an electrically controlled locking mechanism connected to an EV charging socket prevents an EV charger from becoming jammed in the EV charging socket when removal of the EV charger is attempted before the electrically controlled locking mechanism is completely disengaged after charging, in accordance with various embodiments. In diagram **610**, charging has ended and electrically controlled actuator **221** has been deactivated, causing shaft **224** to move upward and away from plunger **222**. However, a user has attempted to remove EV charger **230** from EV charging socket **210** before plunger **222** has fully disengaged from notch **231**. Conventionally, such an attempt can cause a jam.

[0050] In various embodiments, however, tip or pin portion **225** of plunger **222** that comes in contact with notch **231** is chamfered or angled to prevent a jam. A jam is prevented because, when EV charger **230** is pulled from EV charging socket **210**, an edge of notch **231** contacts angled tip **225** causing plunger **222** to move upward. In other words, if the edge of notch **231** and the tip of plunger **222** are parallel flat surfaces, a jam could be created when the parallel surfaces come

together when the removal of EV charger **230** is attempted. The force applied by the removal of EV charger **230** against the parallel surfaces may produce enough perpendicular frictional force between the parallel surfaces to counteract the upward force produced on plunger **222** by one or more springs **223**.

[0051] However, because plunger **222** includes angled tip **225**, when the flat edge of notch **231** contacts angled tip **225**, less perpendicular frictional force is produced. Thus, the upward force produced on plunger **222** by one or more springs **223** can overcome the frictional force, causing angled tip **225** to be moved upward.

[0052] In diagram **620**, the upward force produced on plunger **222** by one or more springs **223** has completely overcome the frictional force produced between the flat edge of notch **231** and angled tip **225**. As a result, plunger **222** has been completely removed from the channel of EV charging socket **210**. EV charger **230** is, therefore, able to be freely moved out of EV charging socket **210**. Thus, the combination of one or more springs **223** and angled tip **225** provides an auto return function for plunger **222** and prevents a possible jam.

Rotary Motor Driven Actuator

[0053] In various embodiments, actuator **221** of FIG. **6** is a rotary motor driven actuator.

[0054] FIG. **7** is an exemplary cross-sectional diagram **700** showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator is connected internally to an EV charging socket, in accordance with various embodiments. In FIG. **7**, electrically controlled locking mechanism housing **220** is connected to the outside of EV charging socket **210**. Electrically controlled locking mechanism housing **220** includes the electrically controlled locking mechanism.

[0055] The electrically controlled locking mechanism includes electrically controlled actuator **221**, plunger or lock pin **222**, and one or more springs **223**. Electrically controlled actuator **221** includes rotary motor **721**.

[0056] EV charging socket **210** includes a hole **211** into channel **212**. Hole **211** is located in a wall of channel **212**. The electrically controlled locking mechanism is positioned above hole **211** so that plunger **222** can be extended through hole **211** and into channel **212** of EV charging socket **210**.

[0057] In various embodiments, plunger **222** is connected to actuator **221**. As shown in FIG. **7**, actuator **221** is not activated. When actuator **221** is activated, rotary motor **721** turns moving actuator **221** down and, in turn, moving plunger **222** down through hole **211** and into channel **212**. When actuator **221** is not activated, one or more springs **223** keep actuator **221** in a position above EV charging socket **210** and, in turn, keep plunger **222** above channel **212**.

[0058] In various embodiments, electrically controlled actuator **221** is activated or deactivated using electrical circuitry (not shown). The electrical circuitry detects when the EV charger is charging or not charging. For example, the electrical circuitry can be electrically connected to the terminals of EV charging socket **210** and, therefore, can determine whether or not electric current is being supplied to the terminals of EV charging socket **210**.

[0059] In various embodiments, electrical circuitry **240** can include one or more processors. The one or more processors can be, but are not limited to, controllers, computers, microprocessors, computer systems such as the computer system of FIG. **1**, or any devices capable of sending and receiving control signals or data and analyzing data. In other words, electrically controlled actuator **221** can be activated or deactivated by receiving controlled signals from one or more processors.

[0060] FIG. **8** is an exemplary cross-sectional diagram **800** showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket locks an EV charger during charging, in accordance with various embodiments. In FIG. **8**, EV charger or charging gun **230** has been inserted into channel **212** of EV charging socket **210**. EV charger **230** includes lockable portion or notch **231**.

[0061] To lock EV charger **230** to or in EV charging socket **210**, electrically controlled actuator **221** of the electrically controlled locking mechanism is activated. This causes rotary motor **721** to turn moving actuator **221** downward and, in turn, plunger **222** downward also. Rotary motor **721**, for

example, moves clockwise to move actuator **221** downward. This downward movement of plunger **222** causes one or more springs **223** to be compressed. It also causes plunger **222** to move down through hole **211** of EV charging socket **210**, to engage notch **231** of EV charger **230**, and to lock EV charger **230** to EV charging socket **210**. Note that actuator **221** remains activated throughout charging in order to keep EV charger **230** locked to EV charging socket **210**.

[0062] FIG. **9** is an exemplary cross-sectional diagram **900** showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket unlocks an EV charger after charging, in accordance with various embodiments. In FIG. **9**, electrically controlled actuator **221** has been deactivated in order to unlock EV charger **230**. The deactivation causes rotary motor **721** to move counterclockwise.

[0063] The deactivation of actuator **221** allows one or more springs **223** to move actuator **221** and, in turn, plunger **222** to move upward and out of notch **231** of EV charger **230**. The movement of plunger **222** upward through hole **211** and out of the channel of EV charging socket **210** unlocks EV charger **230** from EV charging socket **210** and allows EV charger **230** to be removed from socket **210**.

[0064] FIG. **10** is series **1000** of exemplary cross-sectional diagrams showing how an electrically controlled locking mechanism that includes a rotary motor driven actuator and is connected to an EV charging socket prevents an EV charger from becoming jammed in the EV charging socket when removal of the EV charger is attempted before the electrically controlled locking mechanism is completely disengaged, in accordance with various embodiments. In diagram **1010**, charging has not ended. However, a user has attempted to remove EV charger **230** from EV charging socket **210** before plunger **222** is disengaged from notch **231**. This causes plunger **222** and notch **231** to be in contact. After charging has ended, this contact can result in a jam due to the contact between plunger **222** and notch **231**.

[0065] Diagram **1020** shows that after charging contact between plunger **222** and notch **231** can result in a jam. The contact between plunger **222** and notch **231** produces enough frictional force to counteract the pressure produced by one or more springs **223**. In other words, one or more springs **223** cannot move plunger **222** upward.

[0066] Also, if rotary motor **721** is left in the activated position, it also produces additional pressure preventing the jam from being cleared. In various embodiments, however, to allow a user to clear the jam, rotary motor **721** moves counterclockwise, as shown in diagram **1020**, back to the deactivated position. This relieves the additional pressure of rotary motor **721**. Now the user can clear the jam by simply moving EV charger **230** forward. This releases the pressure and one or more springs **223** are able to move plunger **222** upward and out of EV charging socket **210**, as shown in FIG. **9**.

Locking System

[0067] Returning to FIG. **4**, the electrically controlled EV charger locking system includes plunger **222**, one or more springs **223**, and electrically controlled actuator **221**. Plunger **222** physically contacts lockable portion **231** of EV charger **230** inserted into a channel of EV charging socket **210**. When uncompressed, one or more springs **223** position tip **225** of plunger **222** in a return position or location in or above hole **211** of the channel of socket **210**. When compressed, one or more springs **223** apply a force in a second direction to return tip **225** to the return position.

[0068] When activated, electrically controlled actuator **221** applies a mechanical force to plunger **222** in a first direction, which is opposite the second direction, to move tip **225** through hole **211** and to position tip **225** in a locking position in the channel to contact lockable portion **231** of charger **230**. When deactivated, electrically controlled actuator **221** removes the mechanical force from plunger **222**, allowing plunger **222** to be returned to the return position by one or more springs **223**.

[0069] In various embodiments, tip **225** is angled to prevent parallel surface contact between tip **225** and lockable portion **231** of charger **230**.

[0070] In various embodiments, lockable portion **231** includes a notch or slit.

[0071] In various embodiments, plunger **222**, one or more springs **223**, and electrically controlled actuator **221** are enclosed in housing **220** connected to socket **210**.

[0072] In various embodiments, electrically controlled actuator **221** includes a solenoid or rotary motor.

[0073] In various embodiments, electrically controlled actuator **221** receives an electrical signal to activate and an electrical signal to deactivate from electrical circuitry **240**.

[0074] In various embodiments, electrical circuitry **240** includes one or more processors.

[0075] In various embodiments, the electrical signal to activate is received when electrical circuitry **240** detects that charger **230** is charging socket **210**.

[0076] In various embodiments, the electrical signal to deactivate is received when electrical circuitry **240** detects that charger **230** is not charging socket **210**.

Locking Method

[0077] FIG. **11** is an exemplary flowchart showing a method **1100** for locking in EV charger in an EV charging socket, in accordance with various embodiments.

[0078] In step **1110** of method **1100**, an electrical signal is received to activate using an electrically controlled actuator.

[0079] In step **1120**, a mechanical force is applied in a first direction to a plunger using the electrically controlled actuator to move a tip of the plunger through a hole in a channel of an EV charging socket to position the tip in a locking position in the channel to contact a lockable portion of an EV charger inserted into the channel.

[0080] While the present teachings are described in conjunction with various embodiments, it is not intended that the present teachings be limited to such embodiments. On the contrary, the present teachings encompass various alternatives, modifications, and equivalents, as will be appreciated by those of skill in the art.

[0081] Further, in describing various embodiments, the specification may have presented a method and/or process as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims. In addition, the claims directed to the method and/or process should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the sequences may be varied and still remain within the spirit and scope of the various embodiments.

Claims

1. An electrically controlled electric vehicle (EV) charger locking system, comprising: a plunger for physically contacting a lockable portion of an EV charger inserted into an EV charging socket; one or more springs that, when uncompressed, position a tip of the plunger in a return position in or above a hole in a channel of the socket and, when compressed, apply a force in a second direction to return the tip to the return position; and an electrically controlled actuator that, when activated, applies a mechanical force to the plunger in a first direction, which is opposite the second direction, to move the tip through the hole and to position the tip in a locking position in the channel to contact the lockable portion of the charger and, when deactivated, removes the mechanical force from the plunger, allowing the plunger to be returned to the return position by the one or more springs.
2. The system of claim 1, wherein the tip is angled to prevent parallel surface contact between the tip and the lockable portion of the charger.
3. The system of claim 1, wherein the lockable portion comprises a notch.

4. The system of claim 1, wherein the plunger, the one or more springs, and the electrically controlled actuator are enclosed in a housing connected to the socket.
 5. The system of claim 1, wherein the electrically controlled actuator comprises a solenoid or a rotary motor.
 6. The system of claim 1, wherein the electrically controlled actuator receives an electrical signal to activate and an electrical signal to deactivate from electrical circuitry.
 7. The system of claim 6, wherein the electrical circuitry comprises one or more processors.
 8. The system of claim 6, wherein the electrical signal to activate is received when the electrical circuitry detects that the charger is charging the socket.
 9. The system of claim 6, wherein the electrical signal to deactivate is received when the electrical circuitry detects that the charger is not charging the socket.
 10. A method for locking an electric vehicle (EV) charger in an EV charging socket, comprising: receiving an electrical signal to activate using an electrically controlled actuator; applying a mechanical force in a first direction to a plunger using the electrically controlled actuator to move a tip of the plunger through a hole in a channel of an EV charging socket to position the tip in a locking position in the channel to contact a lockable portion of an EV charger inserted into the channel.
 11. The method of claim 10, wherein one or more springs, when uncompressed, position the tip of the plunger to a return position in or above the hole in the channel and, when compressed, apply a force in a first direction to return the tip to the return position.
 12. The method of claim 11, further comprising receiving an electrical signal to deactivate using the electrically controlled actuator, removing the mechanical force from the plunger using the electrically controlled actuator, allowing the plunger to be returned to the return position by the one or more springs.
 13. The method of claim 10, wherein the tip is angled to prevent parallel surface contact between the tip and the lockable portion of the charger.
 14. The method of claim 10, wherein the lockable portion comprises a notch.
 15. The method of claim 11, wherein the plunger, the one or more springs, and the electrically controlled actuator are enclosed in a housing connected to the socket.
 16. The method of claim 10, wherein the electrically controlled actuator comprises a solenoid or a rotary motor.
 17. The method of claim 12, wherein the electrically controlled actuator receives the electrical signal to activate and the electrical signal to deactivate from electrical circuitry.
 18. The method of claim 17, wherein the electrical circuitry comprises one or more processors.
 19. The method of claim 17, wherein the electrical signal to activate is received when the electrical circuitry detects that the charger is charging the socket.
 20. The method of claim 17, wherein the electrical signal to deactivate is received when the electrical circuitry detects that the charger is not charging the socket.
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